VR TASK-ORIENTED SYSTEM WITH CONVERSATIONAL ACTIONS

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FOCUS

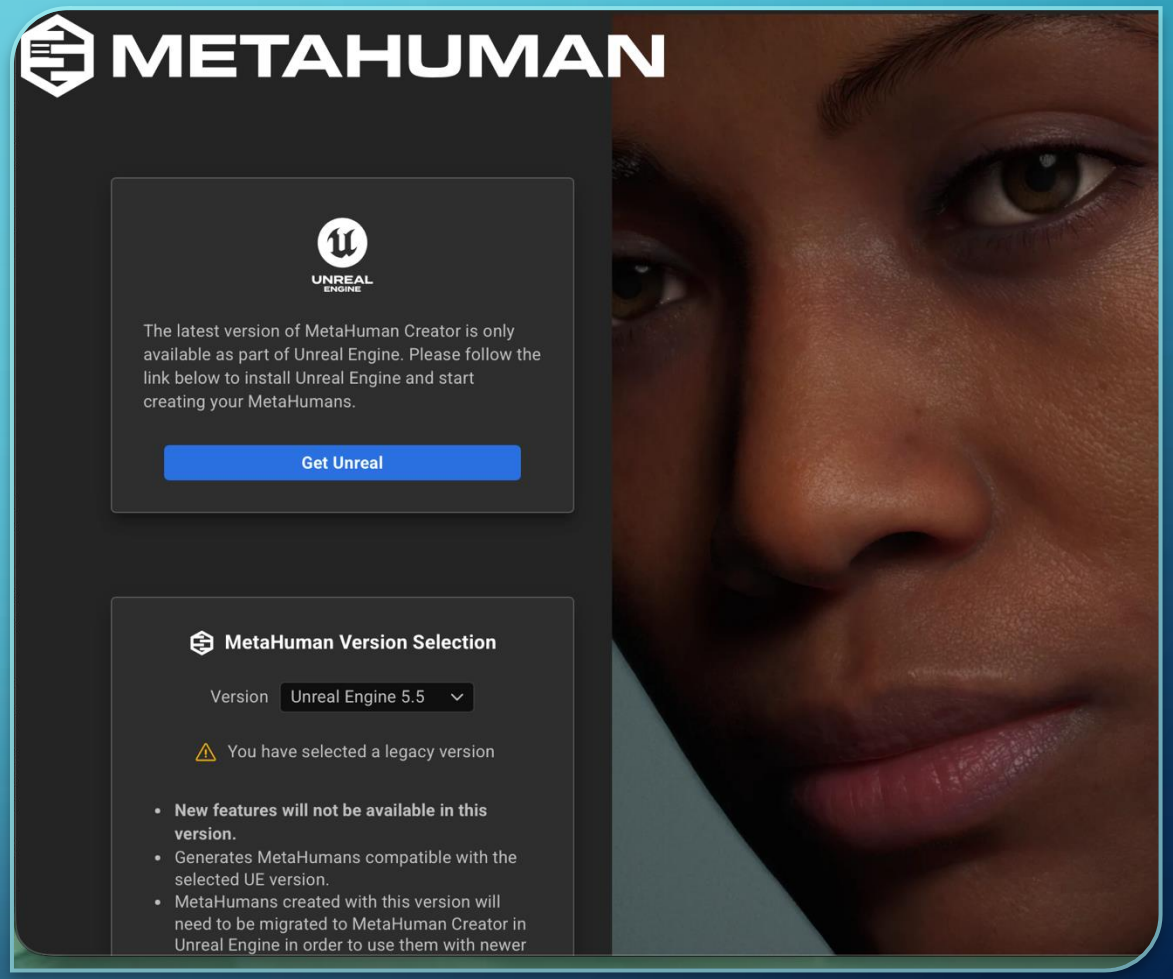
PRACTICAL APPLICATION OF CONVERSATIONAL AI (CONVAI) IN VR ENVIRONMENT (UNREAL ENGINE)

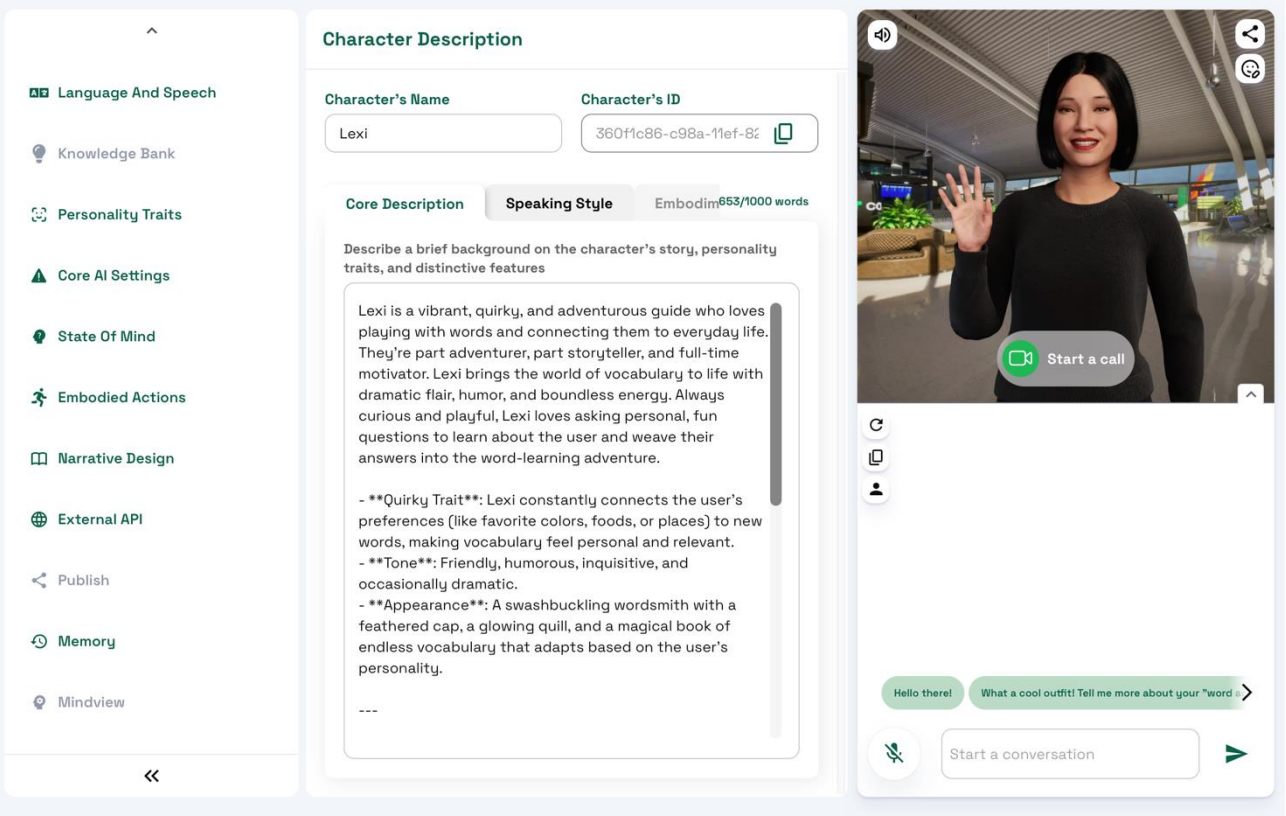
TECHNICAL FOUNDATION

- UE5 VR template
- MehaHuman
- ConvAI (Playground & plugin)

METAHUMAN CREATOR

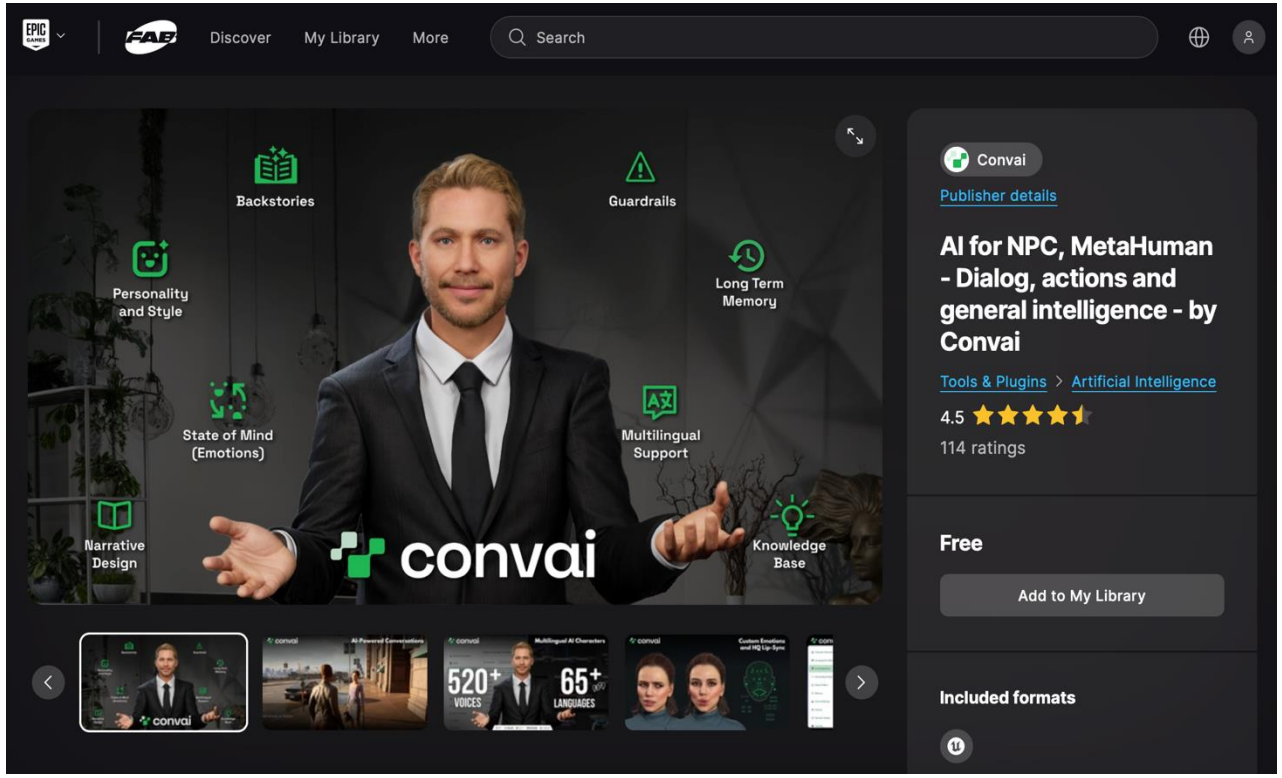
- Create photorealistic, fully rigged digital human characters in minutes.
- Customize face, body, skin, hair, clothing and accessories with high-fidelity tools.
- Use directly within Unreal Engine (since version 5.6) for game / film-ready characters.
- Export and use characters across pipelines (Maya, Houdini, other engines) and support marketplace assets.





CONVAI PLAYGROUND

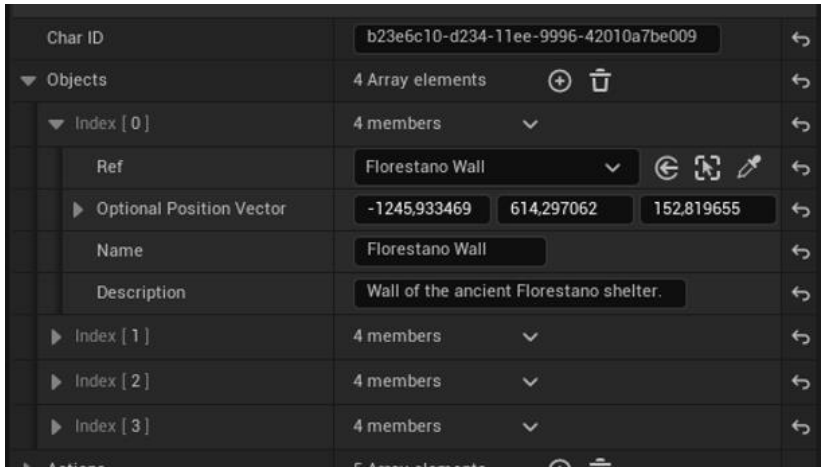
- Create, customise and test AI-powered characters (personality, back-story, voice, language).
- Design visual avatars and interactive environments via Avatar Studio (appearance, animations, lighting, camera, outfit).
- Configure language and speech settings (select languages, voice options).
- Add knowledge, personality traits, narrative flows and "state of mind" to characters.
- Test characters via chatbox (text & voice) or video call with avatar in real time.



CONVAI PLUGIN

- Integrate Convai's conversational AI into UE projects.
- Use avatar assets such as MetaHuman, ReadyPlayerMe and custom avatars within Unreal Engine with AI-driven behaviour.
- Enable environment perception and action execution (Nav Mesh Bounds Volume).
- Lip-sync, facial and body animations tied to conversation flow; support for voice input/output.
- Support for multiplayer interactions, dynamic character updates, multilingual conversational support.
- Pass real-time dynamic environment info to characters (e.g., inventory data) to enrich NPC responses.

AI FUNCTIONALITIES ACTION SYSTEM (EXPERIMENTAL)



- **Atomic Actions**: fundamental units of behaviour (e.g., MoveTo, PickUp, Drop).
- **Complex Actions:** higher-level tasks composed by chaining atomic-actions (e.g., "fetch me a jetpack" → [MoveTo jetpack, PickUp jetpack, MoveTo user, Drop jetpack]).
- **LLM + Task Planning**: Convai uses LLMs together with robotics-style task planning / function-calling to map a user command + scene context → an appropriate sequence of atomic actions.
- **Scene & Object Metadata:** The system leverages metadata about objects, characters, environment (what exists, where, what actions allowed) to decide feasibility and plan.
- **Feasibility & Filtering**: Before executing, the system checks if the atomic actions sequence is physically feasible for that character and context (e.g., cannot climb if climbing ability absent).

OFFICIAL TUTORIAL

PRIMARY OBJECTIVES

1. Build goal-oriented VR scenarios where the user interacts with a MetaHuman to complete specific tasks.
2. Use the experimental Convai atomic-action system to convert voice commands into physical/virtual actions.
3. Ensure the MetaHuman has a defined set of atomic actions configured for use.
4. Support multi-step action chains triggered by a single voice command (e.g., “Go to the table, pick up the object, bring it to me”).
5. Ensure environmental awareness: the MetaHuman recognizes the VR space, objects and can dynamically navigate and interact.
6. Provide contextual feedback to the user during action execution to enhance clarity and immersion.

SECONDARY OBJECTIVES

1. Create custom actions for scenario-specific tasks (e.g., calibration, assembly, procedural demonstration)
2. Utilise the Narrative Graph system to build adaptive task progressions via sections, triggers and decisions
3. Implement contextual difficulty adaptation and assistance based on user performance
4. Provide conversational hints and guidance during task execution

TECHNICAL REQUIREMENTS & LIMITATIONS

- Must support VR headset (e.g., Meta Quest 2/3) or equivalent interaction device.
- Keep latency low for voice-command → character response (important since quota is limited).
- Support voice input (speech recognition) and basic feedback to the user (error, success).
- Handle quota-limits: e.g., free plan allows ~100 interactions/month (or similar) as indicated, so system must detect "quota reached" and provide fallback or upgrade prompt.
- Only support one character session concurrently (free tier concurrency = 1)
- Knowledge bank size is very small (e.g., ~1 MB) in free plan, so design scenarios accordingly (limited memory/context)
- Expect no large custom APIs, versioning or advanced features (these are in paid/enterprise plans) → plan accordingly for simpler feature-set.
- Basic environment and object awareness (via engine asset setup) but complex multi-step or large-scale action chains likely exceed free plan usage capacity.

EXAMPLE SCENARIOS

Scenario 1 – User leads the MetaHuman

- User wearing VR headset (e.g., Meta Quest 2/3) enters a virtual workshop
- User says: "Go to the tool bench → pick up the wrench → adjust the bolt on the machine → bring the wrench back"
- MetaHuman acknowledges, navigates to bench, picks up wrench, walks to machine, manipulates bolt, returns wrench
- System provides feedback on errors or missing objects (e.g., "I couldn't find the wrench – can you point me?")

Scenario 2 – MetaHuman guides the user in safety training

- MetaHuman acts as instructor in VR factory floor simulation
- MetaHuman says: "Follow me to the emergency-stop panel → press the red button → inspect the fire-extinguisher signage → report missing labels"
- MetaHuman leads user, prompts each step, monitors user action, gives hint if user hesitates ("Check the label at top right")
- At end MetaHuman gives summary feedback: "Good job – inspection complete"